CSCI 5521: Machine Learning Fundamentals (Spring 2022)[1]

Sample Questions

1. In I-III, select the correct option(s) (it is not necessary to explain).

I. Select all the option(s) that are activation functions:
(a) Sigmoid (b) Gini index (c) Hyperbolic tangent (d) Entropy (e) Lagrangian

II. Select all the option(s) of feature selection strategy to split a decision tree:
(a) Entropy (b) Lagrangian (c) Gini index (d) Misclassification error (e) Kernel

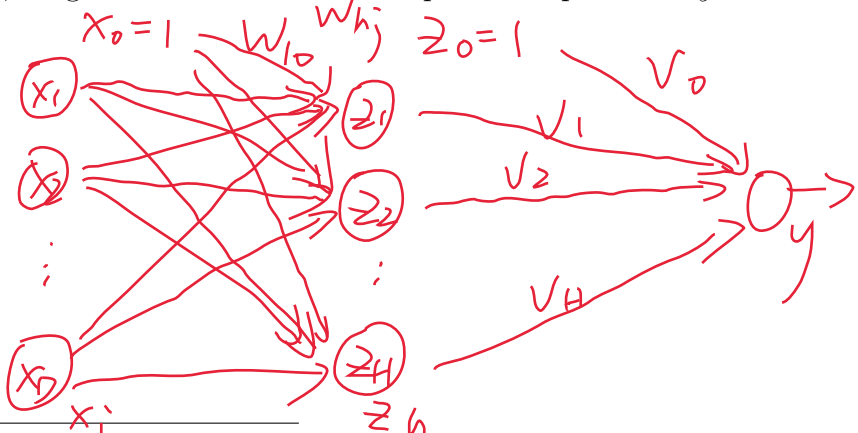
III. Select all the option(s) to control overfitting in decision trees:
(a) Pruning (b) Entropy Threshold (c) Lagrangian (d) Number of leaves (e) Kernel

2. Consider the regularized Multilayer Perceptron (MLP) for binary classification, where each hidden unit uses a sigmoid function. The error function is given by the following expression:

$$E(W, v|X) = -\sum_{t=1}^{N}[r^t \log y^t + (1 - r^t) \log(1 - y^t)] + \frac{\lambda}{2}\sum_{h=1}^{H}||w_h||_2^2 + \frac{\sigma}{2}||v||_2^2$$

where λ and σ are hyperparameters. Answer the following questions:

(a) Draw the multilayer perceptron showing: input values $x_0, ..., x_D$, output of hidden units $z_0, ..., z_H$, weights W and v and the output of output unite y.



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(b) Derive the Forward Step equation for y.

$$z_h^t = \text{sigmoid}\left(\sum_{j=1}^{D} w_{hj} x_j^t + w_{h0}\right)$$

$$y^t = \text{sigmoid}\left(\sum_{h=1}^{H} v_h z_h^t + v_0\right)$$

(c) Derive the Backward Step equations for w_{hj} and v_h.

binary classification (y^t for $p(C_1|x^t)$, $1-y^t$ for $p(C_2|x^t)$)

$$E = E_1 + E_2 + E_3$$

$$\frac{\partial E_1}{\partial w_{hj}} = -\sum_t (r^t - y^t) v_h z_h^t (1 - z_h^t) x_j^t$$

$$\frac{\partial E_2}{\partial w_{hj}} = \lambda w_{hj}$$

$$\frac{\partial E_3}{\partial w_{hj}} = 0$$

$$\Delta w_{hj} = -\eta \frac{\partial E}{\partial w_{hj}} = \eta \left[\sum_t (r^t - y^t) v_h z_h^t (1 - z_h^t) x_j^t - \lambda w_{hj}\right]$$

Similarly, $\Delta v_h = \eta \left[\sum_t (r^t - y^t) z_h^t - \alpha v_h\right]$

3. Build a (Multilayer) Perceptron that calculates a Boolean function (*e.g.*, AND, OR, XOR, NOT, NAND) of its input(s).

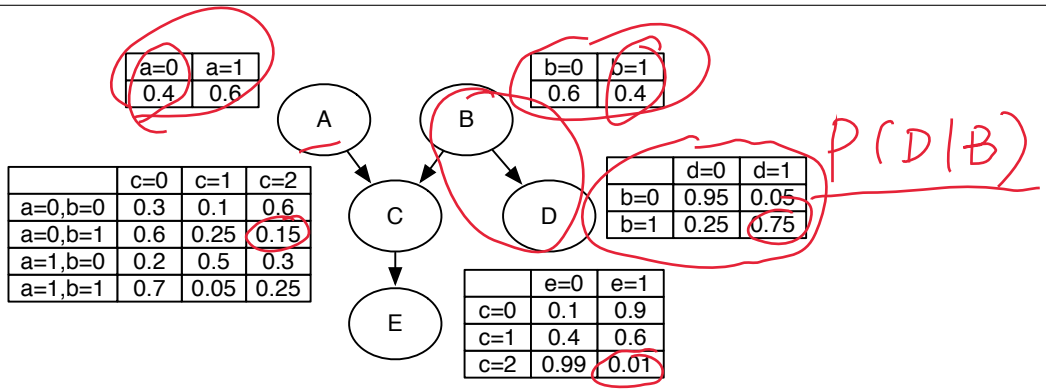
AND $w = [-1.5, 1, 1]^T$

OR $w = [-0.5, 1, 1]^T$

NAND $w = [1.5, -1, -1]^T$

NOT x $w = [0.5, -1]^T$ (input is $[1, x]^T$)

XOR $W = \begin{bmatrix} -0.5 & -0.5 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}$ $v = [0.5, 1, 1]^T$



4. Consider the above Bayesian Network:

   (a) Find the joint probability $P(A, B, C, D, E)$ as the product of conditional probabilities, according to the graphical model given above.

$$P(A,B,C,D,E) = P(A)P(B)P(C|A,B)P(D|B)P(E|C)$$

$$P(A,B,C,D,E) = P(A)P(B|A)P(C|A,B)P(D|A,B,C)P(E|A,B,C,D)$$

   (b) Which of the following statements are true? You can select more than one.

   i. $A \perp\!\!\!\perp B|C$ $\neq$ $A \perp B$

   ii. $C \perp\!\!\!\perp D|B$

   iii. $A \perp\!\!\!\perp B$

   iv. $C \perp\!\!\!\perp D$

$D \perp \{A, C\} | B \iff D \perp A | B$, $D \perp C | B$

$E \perp \{A, B, D\} | C \iff E \perp A | C$, $E \perp B | C$, $E \perp D | C$

   (c) Show how to find the marginal probability $P(A, D)$

$$P(A,D) = \sum_B P(A,B,D) = \sum_B P(A)P(B)P(D|B)$$

$$= P(A) \sum_B P(B)P(D|B)$$

(d) Using the conditional probability distribution (CPD) tables in the figure, find:

i. $P(a = 0, b = 1, c = 2, d = 1, e = 1)$

ii. $P(a = 0, d = 1)$

iii. $P(b = 1, c = 2, e = 1 | a = 0, d = 1)$

$P(A, B, C, D, E)$

i) $P(a=0)\, P(b=1)\, P(C=2 \mid a=0, b=1)\, P(d=1 \mid b=1)\, P(e=1 \mid C=2,$

$= 0.4 \times 0.4 \times 0.15 \times 0.75 \times 0.01 = 1.8 \times 10^{-4}$

ii) $P(A, D)$

$P(a=0) \sum_B P(B)\, P(d=1 \mid B) = P(a=0)\,(P(b=0)P(d=1 \mid b=0)$
$+ P(b=1)P(d=1 \mid b=1))$

$= 0.132$

iii) $\dfrac{P(a=0, b=1, c=2, d=1, e=1)}{P(a=0, d=1)} = 1.32 \times 10^{-3}$